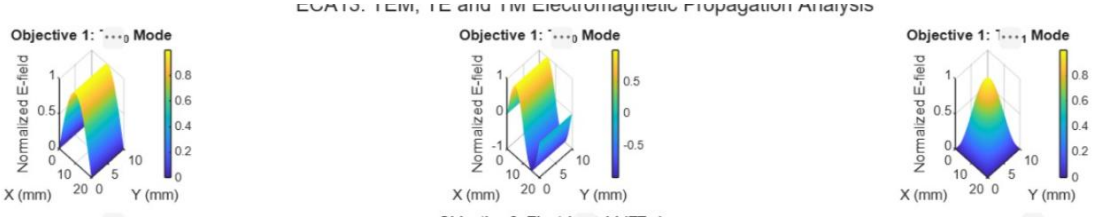
EXPERIMENT 1

Simulation-Based Electromagnetic Analysis of TEM, TE and TM Propagation Modes in Microwave Transmission Structures

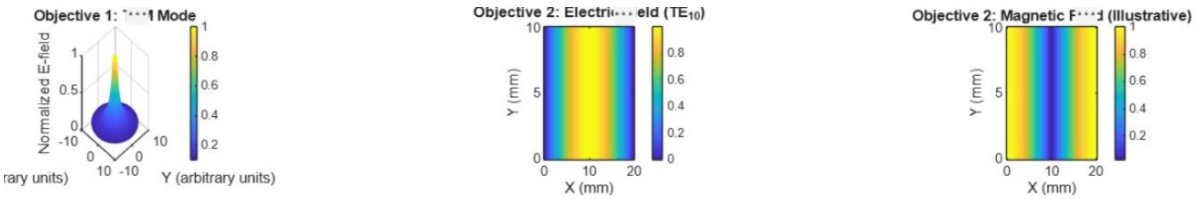
OBJECTIVE 1:

To develop electromagnetic simulation models for TEM, TE and TM propagation modes in suitable microwave transmission structures.



OBJECTIVE 2:

To visualize and analyze the electric-field and magnetic-field distributions associated with different propagation modes.



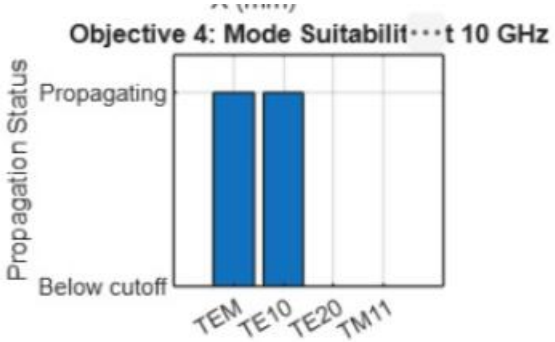
OBJECTIVE 3.

To determine the cutoff frequency, propagation constant and field characteristics of selected TE and TM modes.



OBJECTIVE 4.

To compare the propagation behavior of different modes and identify their suitability for practical microwave transmission.



OUTPUT:

```
========================================
 MICROWAVE ENGINEERING - EXPERIMENT 1
========================================
Operating Frequency = 10.00 GHz

CUTOFF FREQUENCIES
TEM  = 0 GHz (ideal two-conductor line)
TE10 = 7.500 GHz
TE20 = 15.000 GHz
TM11 = 16.771 GHz

PROPAGATION CONSTANTS AT 10.00 GHz
TEM  beta = 209.440 rad/m
TE10 beta = 138.531 rad/m
TE20 is below cutoff at 10.00 GHz
TM11 is below cutoff at 10.00 GHz

PROPAGATION STATUS AT 10.00 GHz
TE10: Propagating
TE20: Below cutoff
TM11: Below cutoff
```

RESULT:

The MATLAB simulation analyzes the propagation characteristics of TEM, TE, and TM modes in electromagnetic waveguides. The 3D surface plots represent the normalized electric field distributions for different propagation modes, highlighting their field patterns and variations across the waveguide. The cutoff-frequency graphs identify the minimum frequency required for TE and TM modes to propagate, while the propagation-constant plots illustrate the relationship between operating frequency, waveguide dimensions, and wave propagation. The results demonstrate the differences between the propagation modes and help in understanding their electromagnetic field distributions and operating characteristics.